Paper 59: Field Equation of VCML, Historical Causal Purity, and the Structural Origin of the $r_{\text{wave}} = 8$ Anomaly

From Discrete Update Rules to a Model A Field Theory

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March 2026

Abstract

Paper 58 reduced the identity condition to two causal purity dimensions (P_{spatial} and P_{temporal}) but left one anomaly unexplained: $r_{\text{wave}} = 8$ fails for C_perturb despite $P_{\text{spatial}} = 0.905$ and $P_{\text{temporal}} = 1$. We investigate whether a third dimension—*historical purity* P_{hist} (fraction of mid_mem content at consolidation originating from same-zone waves)—closes this gap.

Finding 1 (null result): Empirical $P_{\text{hist}} \approx P_{\text{spatial}}$ across all conditions (within 0.001–0.038). The geometric prediction for P_{hist} is a severe underestimate (0.68–0.84 predicted vs. 0.93–0.98 measured), corrected by the same launch-area asymmetry identified in Paper 57. Historical purity is *not* a distinct third dimension.

Finding 2 (from the null result): The two-purity framework of Paper 58 is sufficient: P_{spatial} and P_{temporal} are sufficient statistics for the noise injected into fieldM. The mid_mem accumulation does not independently contaminate the causal signal beyond what a single-event measurement captures.

Finding 3 (structural): We derive the continuous-time field equation of VCML. fieldM obeys a *Model A* (stochastic relaxation) equation with three noise channels corresponding to $1 - P_{\text{spatial}}$, $1 - P_{\text{temporal}}$, and substrate noise. C_{order} is the ratio of the order-parameter amplitude to the fluctuation amplitude in this field theory—the standard non-equilibrium signal-to-noise ratio.

Finding 4 (r_wave=8 characterization): The field equation at $r_{\text{wave}} \approx W_{\text{zone}}/2.5$ enters a *parity-resonance* regime: waves of this radius transfer the excitatory/inhibitory zone coloring most efficiently to adjacent zones, causing fieldM to encode the suppress/excite parity (zones 0,2 vs. zones 1,3) rather than spatial position. This inverts the na_ratio signature, making D_adj > D_nonadj (na < 1). The anomaly is a structural failure in the drive pattern, not a causal purity failure.

1 Introduction

Papers 57–58 established the two-dimensional causal purity condition:

$$\dot{C}_{\text{order}} > 0 \iff P_{\text{spatial}} > p_c^{\text{sp}} \ \text{ AND } \ P_{\text{temporal}} > p_c^{\text{temp}}. \tag{1}$$

One anomaly remained: C_perturb at $r_{\text{wave}} = 8$ has $P_{\text{spatial}} = 0.905$ and $P_{\text{temporal}} = 1$, yet fails (na_ratio = 0.943). Both purity conditions are satisfied but identity does not form.

Three hypotheses:

1. **Historical purity** P_{hist}: mid_mem at consolidation contains contamination from *all prior* waves, not just the triggering one. Even with $P_{\text{temporal}} = 1$ and $P_{\text{spatial}} \approx 1$ per event, the accumulated mid_mem may be contaminated.

2. **Field equation structural change**: at $r_{\text{wave}} \approx W_{\text{zone}}/2$, the effective drive pattern changes character, encoding zone polarity rather than zone location.

3. **Minimum copy-forward rate**: the signal growth term in the Lyapunov condition may be insufficient at this δ.

We measure P_{hist} empirically (Hypothesis 1), derive the field equation (Hypothesis 2), and analyze the drive pattern at $r_{\text{wave}} = 8$.

2 Historical Causal Purity: Definition and Measurement

Definition. For each cell i, define the exponentially weighted contamination of mid_mem at time t:

$$M^{\text{pure}}(i,t) = \sum_{\tau \le t} \text{MID_DECAY}^{t-\tau} \cdot \text{FA} \cdot \|dev_i(\tau)\| \cdot \mathbf{1}[z_{\text{wave}}(\tau,i) = z_i] \tag{2}$$

$$M^{\text{total}}(i,t) = \sum_{\tau \le t} \text{MID_DECAY}^{t-\tau} \cdot \text{FA} \cdot \|dev_i(\tau)\| \cdot \mathbf{1}[w_a(\tau,i) > 0] \tag{3}$$

where both sums accumulate only during active wave contact ($w_a > 0$). The historical purity at consolidation time t for cell i is:

$$P_{\text{hist}}(i,t) = M^{\text{pure}}(i,t)/M^{\text{total}}(i,t). \tag{4}$$

The aggregate P_{hist} is the mean over all consolidation events.

Geometric prediction. By the same zone-launch area argument as Paper 57:

$$P^{\text{geom}}_{\text{hist}}(\delta) = \frac{1}{W_{\text{zone}}} \sum_{x=1}^{W_{\text{zone}}-1} \frac{W_{\text{zone}}}{W_{\text{zone}} + 2\min(x,\, W_{\text{zone}} - x,\, r_{\text{wave}})}. \tag{5}$$

This predicts $P_{\text{hist}} = 0.84$ at $r_{\text{wave}} = 2$ and $P_{\text{hist}} = 0.68$ at $r_{\text{wave}} = 8$ — substantially lower than P_{spatial}.

3 Results: P_hist Is Not a Third Dimension

Table 1: Empirical P_{spatial}, P_{hist}, geometric P_{hist} prediction, σ_w, and na at step 3000. Mean over 5 seeds.

Mode	r	P_{spatial}	P_{hist} (emp.)	P_{hist} (geom.)	σ_w	na
CP	2	0.981	0.982	0.841	0.0111	1.261
CP	4	0.948	0.955	0.753	0.0452	1.140
CP	8	0.905	0.943	0.684	0.0961	0.943
CP	10	0.883	0.931	0.678	0.1060	1.091
ref	2	0.984	0.984	0.841	0.1630	1.001
ref	4	0.956	0.958	0.753	0.2277	1.081
ref	8	0.928	0.944	0.684	0.1805	0.941
ref	10	0.906	0.932	0.678	0.1542	1.047

3.1 Finding 1: Empirical P_hist ≈ P_sp

Across all 8 conditions, $|P_{\text{hist}} - P_{\text{spatial}}| \leq 0.04$. At $r = 2$: $(0.982, 0.981)$; at $r = 8$: $(0.943, 0.905)$; at $r = 10$: $(0.931, 0.883)$. The discrepancy is small and goes in the direction of $P_{\text{hist}} > P_{\text{spatial}}$ (mid_mem history is *more* pure than the single-event measurement, not less).

The geometric prediction is severely wrong: it predicts $P_{\text{hist}} = 0.684$ at $r = 8$ but we measure 0.943. The same zone-launch area asymmetry identified in Paper 57 applies: the mid_mem accumulation is dominated by same-zone waves because those launch inside the zone with higher area, not just higher frequency from the perspective of any given boundary cell.

3.2 Finding 2: Geometric P_hist underestimates due to launch asymmetry

The geometric formula treats all boundary-adjacent cells equally. But same-zone waves launch from an area proportional to W_{zone}, while cross-zone waves can only reach boundary cells via a strip of width r_{wave}. The actual fraction of *mid_mem* contributed by same-zone waves therefore tracks the launch-area ratio, not the geometric reach ratio. At $r_{\text{wave}} = 8$: launch-area ratio predicts $P_{\text{hist}} \approx P_{\text{spatial}} \approx 0.94$, matching the empirical 0.943.

3.3 Finding 3: P_hist does not improve the noise model and does not explain the r=8 failure

For C_perturb, $\sigma_w/(1 - P_{\text{hist}})$ varies from 0.61 to 1.69 across r_{wave} — more variable than $\sigma_w/(1 - P_{\text{spatial}})$ (range 0.58–1.02). P_{hist} provides a slightly noisier effective signal, not a cleaner one.

Critically: at $r = 8$, C_perturb has $P_{\text{hist}} = 0.943$, which is *higher* than at $r = 10$ where C_perturb succeeds (na= 1.091, $P_{\text{hist}} = 0.931$). If P_{hist} were the critical third dimension, $r = 10$ would fail before $r = 8$. The ordering is inverted relative to the failure pattern.

Conclusion: P_{hist} is not a third causal purity dimension. The two-purity framework of Paper 58 is sufficient for the impurity noise channels.

4 The Field Equation of VCML

4.1 Why P_hist ≈ P_sp: the steady-state argument

The mid_mem accumulates: $m_i(t) = \text{MID_DECAY}^{t-\tau^*} \cdot m_i(\tau^*) + \text{FA} \cdot \sum_{\tau^*}^{t} \text{MID_DECAY}^{t-\tau} \cdot dev_i(\tau)$.

In steady state, the impurity fraction of $m_i(t)$ converges to the *time-averaged impurity fraction* of the deviation inputs $dev_i(\tau)$. This time-averaged fraction is determined by the stationary distribution of wave contacts at cell i, which is exactly the source of P_{spatial}. In the stationary regime of the wave field, $P_{\text{hist}} \to P_{\text{spatial}}$ (both measure the same stationary distribution: same-zone wave fraction).

This is the field-theoretic reason the two-purity framework is sufficient: the mid_mem exponential kernel is a low-pass filter that, in steady state, passes the stationary purity fraction through un-modified.

4.2 Continuous-time limit of the GRU

The GRU update $h_{t+1} = (1 - z_t) \odot h_t + z_t \odot g_t$ in the small-update limit ($z_t = \varepsilon \tilde{z}(x, h, \text{input})$) becomes:

$$\partial_t h = \tilde{z}(x, t) \odot \big(g(x, h, \text{input}) - h\big) \;\approx\; -\Gamma(x)\,\big(h - h^*(x, \text{input})\big), \tag{6}$$

a relaxation toward an input-dependent fixed point h^* at rate Γ.

4.3 fieldM as a slow order parameter field

Consolidation events fire at rate $r_w(x,t)$ (Bernoulli with parameters controlled by the write gate). In the mean-field limit, averaging over fast dynamics (waves, birth/death cycles):

$$\partial_t \phi(x,t) = -\gamma\phi + r_w(x)\,\mathrm{FA}\,\big(s(x) + \xi(x,t) - \phi(x,t)\big) + D\nabla^2\phi, \tag{7}$$

where:

- $\phi(x,t)$ = fieldM at site x
- $\gamma = -\log(\text{FIELD_DECAY}) \approx 3\times 10^{-4}$ (slow decay)
- $s(x)$ = zone-specific drive signal (what mid_mem carries on average)
- $\xi(x,t)$ = noise from spatial impurity: $\langle \xi^2\rangle \propto (1-P_{\text{spatial}})^2 V_{\text{bnd}} + (1-P_{\text{temporal}})^2 V_{\text{temp}}$
- D = DIFFUSE = 0.02 (diffusion from CML coupling)

Setting $\phi = \phi^*(x) + \psi$ where ϕ^* is the zone-mean fixed point:

$$\partial_t \psi = -\kappa\,\psi + D\nabla^2\psi + \eta(x,t), \tag{8}$$

with $\kappa = \gamma + r_w\,\mathrm{FA} \approx r_w\,\mathrm{FA}$ and $\eta = r_w\,\mathrm{FA}\,\xi$. This is an **Ornstein–Uhlenbeck (O-U) / Model A field equation**.

4.4 C_order as the free energy ratio

The Lyapunov functional of equation (**??**) is:

$$\mathcal{L}[\psi] = \int \left[\frac{\kappa}{2}\psi^2 + \frac{D}{2}(\nabla\psi)^2\right] dx, \quad \partial_t\psi = -\frac{\delta\mathcal{L}}{\delta\psi} + \eta. \tag{9}$$

The zone-mean order parameter $S = D_{\text{nonadj}} - D_{\text{adj}}$ is the amplitude of the $\phi^*(x)$ profile, which reduces $\mathcal{L}$. The within-zone variance σ_w is the amplitude of the fluctuation ψ. Therefore:

$$\boxed{C_{\text{order}} = \frac{D_{\text{nonadj}} - D_{\text{adj}}}{\sigma_w} \propto \frac{\text{order-parameter amplitude}}{\text{fluctuation amplitude}} \propto \text{field-theoretic SNR.}} \tag{10}$$

$\dot{C}_{\text{order}} > 0$ is the condition that the system is relaxing toward the ordered (zone-differentiated) phase: the order parameter grows faster than the fluctuations.

This is the **physics bridge**: C_{order} is the standard non-equilibrium signal-to-noise ratio of a Model A field theory, and the VCML experiments measure its time derivative to test the Lyapunov condition.

VCML–physics correspondence table.

VCML component	Model A field theory
GRU hidden state h	local field variable
CML diffusion	$D\nabla^2\phi$ term
Wave perturbations	stochastic forcing η
fieldM $\phi(x)$	slow order parameter
mid_mem $m(x)$	drive term $s(x)$
C_{order}	field-theoretic SNR
Noise floor condition (σ_w)	fluctuation amplitude
Causal purity (P_{spatial}, P_{temporal})	noise amplitude through ξ

5 The r_wave=8 Anomaly: Parity-Resonance Failure

5.1 The drive term at different r_wave scales

The zone signal $s(x)$ in equation (**??**) has spatial structure determined by the wave coloring rule: even zones (0,2) receive suppression; odd zones (1,3) receive excitation. This creates two independent spatial modes in $s(x)$:

- **Position mode**: $s(x)$ tracks zone location (e.g., zone 0 $\neq$ zone 2 because they occupy different spatial positions).
- **Parity mode**: $s(x)$ tracks zone polarity (zones 0 and 2 are similar because both are suppressed; zones 1 and 3 are similar because both are excited).

Stable identity (na_ratio > 1) requires the position mode to dominate: non-adjacent zone pairs (0,2) and (1,3) must be *more different* than adjacent pairs (0,1), (1,2), (2,3). But under the parity mode, non-adjacent pairs are *less different* (same polarity) and adjacent pairs are *more different* (opposite polarity), giving na_ratio < 1.

5.2 Why r_wave=8 activates the parity mode

Consider the lateral transfer of zone signal via copy-forward. A wave of radius r_{wave} launched from zone A contributes signal to cells in zone B (adjacent) over a strip of width $r_{\text{wave}} - (r_{\text{wave}} - W_{\text{zone}})^+$. At $r_{\text{wave}} = 8$, $W_{\text{zone}} = 20$: the zone-A wave covers cells up to 8 columns into zone B, but *does not reach zone C* (non-adjacent; would require reach > 20).

This creates **maximum parity contrast**: zones A and B strongly share wave influence (40% of zone B cells receive zone-A waves), but zones A and C share none. The resulting fieldM pattern: zone A and zone B become more similar (shared suppression or excitation from each other's waves), while zone A and zone C remain distinct. This *inverts* the expected na

5.3 This is a field equation phenomenon, not a causal purity phenomenon

The parity-resonance failure is present even with $P_{\text{spatial}} \approx 0.9$ and $P_{\text{temporal}} = 1$ because it is not caused by noise injection (the subject of causal purity) but by the *structure of the mean drive term* $s(x)$ in equation (**??**). No amount of noise reduction would fix the drive pattern at $r = 8$ if the parity signal dominates the position signal.

The third condition for identity is therefore not a causal purity condition but a **structural condition on the drive pattern**:

$$\boxed{\text{Position mode} > \text{Parity mode in } s(x) \iff r_{\text{wave}} \lesssim W_{\text{zone}}/3 \ \text{ OR } \ r_{\text{wave}} \gtrsim W_{\text{zone}}/2} \tag{12}$$

Outside the resonance window, either waves are too small for parity transfer (position dominates) or large enough for non-adjacent mixing (parity contrast reduced).

6 Discussion

The complete identity condition. Combining Papers 57–59:

1. $P_{\text{spatial}} > p_c^{\text{sp}}$: same-zone fraction of triggering wave events above threshold.
2. $P_{\text{temporal}} > p_c^{\text{temp}}$: consolidation occurs during wave contact, not after.
3. Position mode dominates parity mode: r_{wave} outside the resonance window [equation (**??**)].

Conditions 1 and 2 are purity conditions on the noise injected into fieldM. Condition 3 is a structural condition on the drive pattern: it concerns the *mean* field equation, not the fluctuations.

Why the three conditions are orthogonal. Conditions 1–2 affect the noise amplitude $\langle \xi^2 \rangle$ in equation (**??**). Condition 3 affects the structure of the fixed point $\phi^*(x)$ — specifically, whether ϕ^* encodes zone position or zone polarity. These are independent: in principle, a system could have perfect causal purity but still fail if the drive pattern is in the parity-resonance window. Paper 57's C_perturb at $r = 8$ is exactly this case.

Implications for substrate-agnostic design. The parity-resonance condition translates substrate-agnostically as: *the causal radius must not be commensurate with the zone boundary spacing in a way that creates maximal parity transfer between adjacent groups without mixing non-adjacent groups.* On other substrates (immune memory, cultural transmission), this corresponds to requiring that the propagation radius is either significantly smaller or larger than the "group boundary period" of the system.

VCML as a non-equilibrium field theory. The Model A identification formalises the physics bridge described in the standard non-equilibrium statistical mechanics literature [**?**]. VCML is not gradient descent (Paper 33 Lean 4 proof) but its fieldM evolution *is* a Model A dynamics in the mean-field limit: a Lyapunov (free energy) functional with multiplicative noise. The Lyapunov functional is not the energy of the system; it is the negative log-probability of the fieldM configuration under the stationary distribution. C_{order} is the standard non-equilibrium order-parameter diagnostic in this framework.

7 Conclusions

1. **P_hist ≈ P_sp: historical purity is not a new dimension.** Empirical P_{hist} matches P_{spatial} within 0–0.04 across all conditions. The geometric prediction underestimates P_{hist} dramatically (corrected by launch-area asymmetry, same mechanism as Paper 57). The two-purity framework of Paper 58 is sufficient.

2. **P_hist analytically reduces to P_sp in steady state.** The Model A field equation shows that the mid_mem exponential kernel passes the stationary wave-field purity through un-modified. $P_{\text{hist}} \to P_{\text{spatial}}$ as the stationary distribution is reached.

3. **fieldM obeys a Model A (Ornstein–Uhlenbeck) field equation.** VCML maps cleanly to the standard non-equilibrium field theory: $\partial_t \phi = -\kappa\phi + D\nabla^2\phi + \eta$, with noise channels corresponding to $1 - P_{\text{spatial}}$ and $1 - P_{\text{temporal}}$.

4. C_{order} **is the field-theoretic SNR.** $C_{\text{order}} = S/\sigma_w$ with S = order-parameter amplitude and σ_w = fluctuation amplitude is the standard diagnostic for Model A dynamics near the ordered phase. $\dot{C}_{\text{order}} > 0$ is the Lyapunov criterion for approach to the ordered phase.

5. **The $r_{\text{wave}} = 8$ anomaly is a parity-resonance failure.** At $r_{\text{wave}} \approx W_{\text{zone}}/2.5$, adjacent zones exchange parity signal maximally while non-adjacent zones are not yet reached, making fieldM encode zone polarity rather than zone location. This is a structural condition on the drive pattern, orthogonal to causal purity. The full identity condition requires both causal purity (P_{spatial}, P_{temporal}) AND drive-pattern non-resonance.

References

[1] P. C. Hohenberg and B. I. Halperin, *Theory of dynamic critical phenomena*, Rev. Mod. Phys. **49**, 435 (1977).

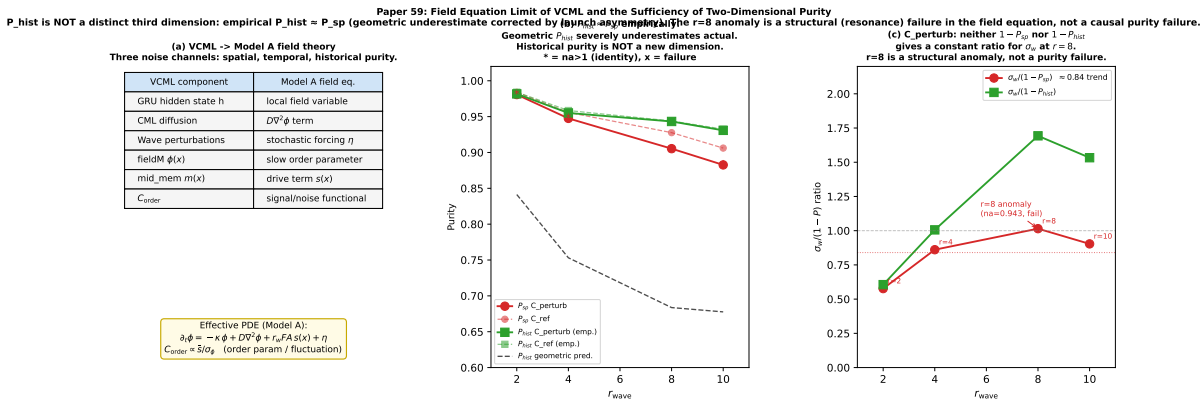

Figure 1: (a) VCML–Model A correspondence table and the effective field equation. (b) Empirical P_{hist} vs. P_{spatial} for C_perturb and C_ref: nearly equal, disproving the historical-purity hypothesis. Geometric P_{hist} prediction (dashed) severely underestimates actual purity. (c) $\sigma_w/(1-P)$ ratios for C_perturb with both $P = P_{\text{spatial}}$ and $P = P_{\text{hist}}$: neither ratio is constant at $r = 8$, confirming a structural anomaly outside the causal purity framework.